extremely short scales, space and spacetime become unavoidably wavy and bumpy. But (Chapter 7), a full assessment of quantum mechanics and time requires a resolution of the quantum measurement problem. One of the proposals for doing so, the Many Worlds interpretation, is particularly relevant for coping with paradoxes arising from time travel, and we will take that up in the next section. But in this section, let's stay classical and bring the block-of-ice/loaf-of-bread depiction of spacetime to bear on these puzzles.

Take the paradoxical example of your having gone back in time and having prevented your parents from meeting. Intuitively, we all know what that's supposed to mean. Before you time-traveled to the past, your parents had met—say, at the stroke of midnight, December 31, 1965,[43] at a New Year's party—and, in due course, your mother gave birth to you. Then, many years later, you decided to travel to the past—back to December 31, 1965—and once there, you changed things; in particular, you kept your parents apart, preventing your own conception and birth. But let's now counter this intuitive description with the more fully reasoned spacetime-loaf depiction of time.

At its core, the intuitive description fails to make sense because it assumes moments can change. The intuitive picture envisions the stroke of midnight, December 31, 1965 (using standard earthling time-slicing), as "initially" being the moment of your parents meeting, but envisions further that your interference "subsequently" changes things so that at the stroke of midnight, December 31, 1965, your parents are miles, if not continents, apart. The problem with this recounting of events, though, is that moments don't change; as we've seen, they just are. The spacetime loaf exists, fixed and unchanging. There is no meaning to a moment's "initially" being one way and "subsequently" being another way.

If you time-traveled back to December 31, 1965, then you were there, you were always there, you will always be there, you were never not there. December 31, 1965, did not happen twice, with your missing the debut but attending the encore. From the timeless perspective of Figure 5.1, you exist—static and unchanging—at various locations in the spacetime loaf. If today you set the dials on your time machine to send you to 11:50 p.m., December 31, 1965, then this latter moment will be among the locations in the spacetime loaf at which you can be found. But